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Can AI Do GRB Data Analysis? A Human-versus-Agent Benchmark on Time-Resolved *Fermi*/GBM Spectroscopy

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ABSTRACT

Can current artificial-intelligence systems carry out the judgement-laden steps of gamma-ray burst (GRB) data analysis? Here we show that they largely can. Time-resolved spectroscopy of GRBs rests on a sequence of expert, partly subjective choices: which detectors to use, where to place the background, which interval contains the source, and how to bin the light curve. We benchmark several leading agentic AI systems—Anthropic's Claude, OpenAI's Codex, and Google's Antigravity—on the analysis of single-pulse *Fermi*/GBM bursts. Each system is given only a set of natural-language instructions, in the form of machine-readable Markdown task guides, together with a standard analysis toolchain, and is asked to execute the entire workflow: retrieving the data, selecting detectors from spacecraft pointing, modeling the polynomial background, defining the source interval, binning the light curve (Bayesian blocks with a significance floor), and fitting time-resolved spectra. A single auditable *approval gate* records, for every selection, who or what made it; because each system reads the *identical* guides, any difference reflects the model and not the instructions, and the pipeline doubles as a controlled human-versus-AI experiment. We score each system against a panel of experienced human analysts—referenced to the inter-expert scatter, so a system is judged successful if it matches humans about as well as humans match each other—and trace any disagreement into the recovered physics ($E_{\rm p}$, low-energy index, blackbody kT, and the thermal-versus-double-break classification). **[PENDING run: Result, pending the benchmark run: each system reproduces expert selections to within the inter-human scatter and leaves the inferred spectral physics statistically unchanged, performing comparably to an experienced researcher.]** We release the fully agent-legible, self-documenting code so the analysis can be re-run end to end by a human or any agent, and discuss the implications for reproducibility, for scaling GRB spectral catalogs to the full single-pulse population, and the regimes in which agents do and do not match expert judgement.

Keywords: Gamma-ray bursts (629) — Astronomy data analysis (1858) — Convolutional neural networks — Astrostatistics (1882)

1. INTRODUCTION

Time-resolved spectroscopy is the primary tool for probing the radiation physics of the GRB prompt phase. In practice, turning raw *Fermi*/GBM event data into a time series of spectral fits requires a chain of human decisions: selecting the illuminated NaI and BGO detectors, defining off-source intervals for the background model, identifying the source/emission interval, and choosing a binning that balances time resolution against per-bin signal. These choices are expert, somewhat subjective, and seldom recorded at the level of detail needed to reproduce a published catalog exactly. As samples grow, this manual bottleneck both limits throughput and undermines reproducibility.

Automating these choices, and validating the automation against people, is not new to GRB science. Busby & Lazzati (2024) classified the morphology and spectral evolution of 62 bright *Fermi*/GBM bursts with two independent methods—a rule-based computational test (a "horizontal-line" single-peak test plus a peak-energy monotonicity test) and a *citizen-science* test in which 24 participants ranked every light curve and spectrum—

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and compared the two with Pearson and Spearman statistics, finding that the computer and human rankings generally agreed (single-peak Spearman $r_s = 0.81$; hard-to-soft $r_s = 0.65$). That algorithm-versus-human comparison on a burst-level *classification* is the direct precedent for this work, and its multi-rater (citizen-science) design motivates using inter-human scatter as the baseline against which automation is judged.

Large-language-model (LLM) "agents"—models that can read a repository, run code, inspect figures, and act on the results—now make it possible to automate not just a fixed classification score but the *open-ended judgements* of the analysis itself **[cite: recent astro-LLM / agent works]**. Their promise for GRB analysis is specific: the subjective steps above are largely visual judgements on a light curve, exactly the kind of task a vision-capable agent can attempt, and the rest of the workflow is deterministic code an agent can orchestrate. We therefore extend the Busby & Lazzati (2024) computer-versus-human paradigm in two ways: from a fixed heuristic computing a scalar score to an *LLM agent making the actual analysis decisions*, and from a single classification label to the *full pipeline* with its downstream effect on the physics. The open question is not whether an agent *can* run the pipeline, but whether its judgements match an expert's well enough that the science is unchanged—and whether the process can be made auditable rather than a black box.

In this paper we (1) describe an agentic pipeline for single-pulse GBM GRBs built around a human-or-AI *approval gate* that makes every selection auditable; (2) use that gate to benchmark agent against expert on the subjective steps and measure the downstream impact on spectral parameters; and (3) validate the pipeline by reproducing known single-pulse results. The single-pulse spectroscopy science itself is presented in a companion paper (Sharma et al., in prep.); here the focus is the *method* and its validation. Section 2 describes the sample and data; Section 3 the agentic pipeline; Section 4 the human–AI experiment; Section 5 the results; and Sections 6–7 the discussion and conclusions.

2. SAMPLE AND DATA

We analyze the single-pulse *Fermi*/GBM sample of the companion science paper: bursts selected by the shape-based "horizontal-line" criterion of Busby & Lazzati (2024) (a fast-rise/exponential-decay single-pulse score), with a fluence cut to ensure adequate time-resolved signal-to-noise. The selection is on pulse *shape*, not brightness. We use the time-tagged event (TTE) data of the NaI detectors within $\sim 50^\circ$ of the source, the most-illuminated BGO, and, where available, LAT Low-Energy (LLE) data. For the human–AI benchmark (Section 4) we additionally define a subset of N_{bench} = **[PENDING run: 20-30]** bursts processed in *both* approval modes.

3. THE AGENTIC PIPELINE

The pipeline is a self-documenting code repository that an AI coding agent reads and executes. A top-level machine-readable guide (an `AGENTS.md` file) orients any agent—human-invoked or autonomous—to the environment, data, run order, and the approval semantics, so a fresh agent with no prior context can drive the full chain. The analysis proceeds in three gated stages.

3.1. *Stage 1: gated selection (detectors, background, source)*

The selections that are traditionally manual are unified into one step that writes a single, stamped catalog. For each burst the agent (or a human) approves:

1. *Detectors.* Detector–source angles are computed from the spacecraft pointing (POSHIST quaternions); NaI detectors within a threshold angle plus the matching BGO are proposed and approved.
2. *Background.* Pre- and post-burst off-source windows are fit with a time polynomial (order chosen by a likelihood-ratio test) and interpolated beneath the burst, following standard GBM practice.
3. *Source.* The emission interval is marked explicitly per burst on the brightest-detector light curve.

Crucially, the approval is gated: every catalog row carries an *approval stamp* recording the approver and the mode—`human_gui` (a person marks the windows interactively) or `ai_vision` (the agent renders the light curves, inspects the images, and proposes the windows). An agent thus never silently substitutes its judgement for a human's; the provenance of every decision is recoverable from the output file. This stamp is the foundation of the benchmark in Section 4.

3.2. *Stage 2: adaptive binning*

The light curve is partitioned with the Bayesian Blocks algorithm (Scargle et al. 2013) computed on the background-subtracted rate, combined with a per-block significance floor: blocks below the floor are merged so that every surviving spectral bin clears a detection threshold. The binning runs within the approved source interval.

3.3. *Stage 3: spectral fitting and model selection*

Each bin is fit with a family of empirical and photospheric models (**[cite: Band 1993, CPL, SBPL/Ravasio, DSBPL, Band+BB, CPL+BB]**) using a forward-folding likelihood with the detector responses and the polynomial background propagated as Gaussian errors (the `pgstat` statistic). The preferred model per bin is chosen by an information criterion with a physical-validity gate. **[cite: 3ML / astromodels (Vianello et al.)]**

3.4. *Built-in adversarial verification*

A distinguishing feature of the agentic implementation is that verification is itself automated: code changes are reviewed by an ensemble of independent agents that attempt to refute each other's findings before a change is accepted. We report this protocol as part of the method (Appendix B), since it materially affects the reliability of an agent-built pipeline.

4. EXPERIMENTAL DESIGN: HUMAN VERSUS AI

The approval gate (Section 3.1) lets us run the subjective steps in both modes on the identical benchmark sample and hold everything downstream fixed. Following Busby & Lazzati (2024), the human baseline is provided by *multiple* independent expert approvers, so that the inter-human scatter sets the scale against which the agent is judged: the agent is consistent with experts if it matches them as well as they match each other. We define two families of metrics.

Systems benchmarked.—We evaluate several contemporary agentic coding systems—e.g. OpenAI Codex, Google Antigravity, and Anthropic Claude Code **[PENDING run: fix exact versions]**—each operating the same repository through the same machine-readable task guides (Section 3). Holding the instructions fixed across systems makes the comparison a test of the *models*, not of the prompts. Each system runs every judgement step (detector, background, source, QC) in `ai_vision` mode and writes its own stamped catalog; the human experts write theirs in `human_gui` mode. We then report, per metric, each system against the inter-expert band (a per-system leaderboard) and the systems against one another (do independent LLMs converge on the same selections?).

Selection agreement.—For each burst we compare each pair of raters: the approved-detector set agreement (Jaccard), the background-window overlap (intersection-over-union) and per-edge differences, and the source-interval edge differences and duration error. For every metric we report the AI-versus-human distribution against the human-versus-human distribution. **[PENDING run: report values.]**

Downstream impact (the decisive test).—We propagate each mode's selections through the identical binning and fitting and compare the recovered quantities bin-by-bin and burst-by-burst: $E_{\rm p}$, the low-energy index, the blackbody kT where a thermal component is preferred, and the thermal-versus-double-break classification. As in Busby & Lazzati (2024) we quantify agreement with Pearson and Spearman statistics (here on the recovered parameters rather than on a classification score), again referenced to the inter-human scatter. The question is whether scientific conclusions are invariant under the human→AI swap. **[PENDING run: report the parameter scatter, correlation, and classification concordance.]**

5. RESULTS

[PENDING run: This entire section awaits the dual-mode run. Planned content:]

- *Fig. 1*: per-burst selection-agreement summary (detector concordance; background- and source-edge differences).
- *Fig. 2*: AI-vs-human scatter of $E_{\rm p}$, low-energy index, and kT with the one-to-one line and intrinsic scatter.
- *Table 1*: concordance of the per-bin model classification.
- *Fig. 3*: a representative burst shown end to end in both modes.
- *Validation*: recovery of established single-pulse relations (e.g. the $E_{\rm p}$–kT correlation) under the agentic pipeline.

6. DISCUSSION

[PENDING run: Frame around the results.] Anticipated axes: where the agent agrees with experts and where it does not (e.g. faint bursts, complex backgrounds, orbital trends); the reproducibility and auditability gains from the stamped gate; the throughput implications for scaling time-resolved catalogs; and an honest account of failure modes and the necessity of the human-in-the-loop option. Positioning relative to other agentic/LLM analyses in astronomy **[cite: related work]**.

7. CONCLUSIONS

[PENDING run: Write after results.] We present and benchmark an auditable agentic pipeline for single-pulse GBM time-resolved spectroscopy; the gated, human-or-AI design makes the traditionally manual choices reproducible and lets us measure, rather than assume, the impact of delegating them to an agent.

APPENDIX

A. REPRODUCIBILITY

The full pipeline, its agent guide (`AGENTS.md`), and the exact commands for each stage are released with this paper. The analysis can be re-run end to end by a human or an AI agent from the released repository. **[cite: repository DOI / URL on publication.]**

B. ADVERSARIAL VERIFICATION PROTOCOL

[PENDING run: Describe the multi-agent review ensemble used to validate the implementation.]

REFERENCES

Busby, I., & Lazzati, D. 2024, ApJ, 972, 83

Scargle, J. D., Norris, J. P., Jackson, B., & Chiang, J. 2013, ApJ, 764, 167